

Project Initialization and Planning Phase

Date	5-7-2026
Team ID	
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve the learning experience using Generative AI by providing instant question answering, concept explanations, quiz generation, text summarization, and personalized learning recommendations. EduGenie helps students learn faster, understand concepts easily, and prepare effectively through an AI-powered web application.

Project Overview	
Objective	Develop an AI-powered learning assistant that provides accurate answers, concept explanations, quizzes, summaries, and personalized learning recommendations to improve students' learning experience.
Scope	The project covers AI-powered educational assistance using Google Gemini, FastAPI, HTML, CSS, Jinja2, and SQLite to support students with interactive learning features through a web application.
Problem Statement	
Description	Students often struggle to understand concepts, revise lengthy notes, create quizzes, and find a proper learning path from multiple online resources
Impact	The solution improves learning efficiency, saves study time, enhances understanding, and helps students prepare confidently for exams.
Proposed Solution	
Approach	Use Google Gemini AI with FastAPI to process user queries and provide intelligent educational assistance through a simple web interface
Key Features	- AI-based Question Answering - Concept Explanation - Quiz Generation - Text Summarization - Personalized Learning Recommendations - Responsive Web Interface

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5 / AMD Ryzen 5 or higher
Memory	RAM specifications	8 GB RAM
Storage	Disk space for project and database	256 GB SSD
Software		
Frameworks	Python framework	FastAPI
Libraries	Additional libraries	google-generativeai, uvicorn, jinja2, python-dotenv, python-multipart
Development Environment	IDE	Visual Studio Code
Data		
Database	Data storage	SQLite Database
AI Model	Generative AI model	Google Gemini 2.5 Flash